

Computer Communication and Networking

Project Report

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1. Executive Summary

⑨ This report presents the complete design and implementation of a network infrastructure for a medium-sized organization using Cisco Packet Tracer. The network is divided into multiple departments, including IT, Finance, Computer Science, and Management, to ensure efficient communication and proper resource utilization. The primary objective of this project is to develop a secure, scalable, and well-structured network that supports dynamic IP address allocation, departmental traffic isolation through VLANs, and reliable connectivity between all network segments. Various networking concepts such as VLAN configuration, DHCP, basic device security, and verification techniques have been implemented to meet the requirements of the Computer Communication and Networking (CCN) Lab.

2. Project Overview

The network for this project has been designed and implemented using Cisco Packet Tracer with the aim of providing reliable communication among different departments within the organization. The design emphasizes secure device configuration, efficient network management, and logical traffic separation. To achieve these objectives, VLAN segmentation has been implemented to isolate departmental traffic, while DHCP services are configured to enable automatic IP address assignment to end devices. In addition, dynamic routing using the RIP protocol is configured to ensure proper communication between different network segments of the network. Configuration backup through a centralized TFTP server is also incorporated to support data recovery and maintain overall network reliability.

3. IP Addressing Scheme

The IP addressing scheme used in this project is based on the last two digits of the student enrollment number, as required in the project guidelines. Instead of assigning individual IP addresses at every location, a structured addressing approach was adopted to maintain simplicity and consistency across the network. The enrollment number digits were incorporated into the addressing scheme and were also used during the RIP routing configuration to uniquely identify and manage network segments. This approach ensures compliance with the given addressing rules while maintaining efficient IP management throughout the network.

4. Network Design Methodology

4.1 Topology Design Rationale

A hierarchical network design model has been implemented in this project to ensure organized communication and efficient network management. The network is divided into three logical layers: **Core Layer:** A central router is used to perform inter-VLAN routing and manage communication between different network segments.

Distribution Layer: Multiple switches are deployed to connect and manage each department separately.

Access Layer: End devices such as PCs, laptops, and servers are connected at this layer to provide user access to the network. This hierarchical design improves network scalability, enhances fault isolation, and provides better overall performance by efficiently managing network traffic.

4.1.1 Topology Selection and Justification:

⑨ A **hybrid topology** is used in this project. Star topology is implemented at the departmental level where end devices are connected to switches. Additionally, the routers are interconnected in a **ring topology**, forming a closed loop to ensure continuous network connectivity. This combination improves reliability, allows efficient communication between departments, and supports future network expansion.

4.2 Device Configuration

All routers and switches in the network were configured with essential system and security settings to ensure stable and secure operation. These configurations help in proper device identification, secure access control, and smooth network

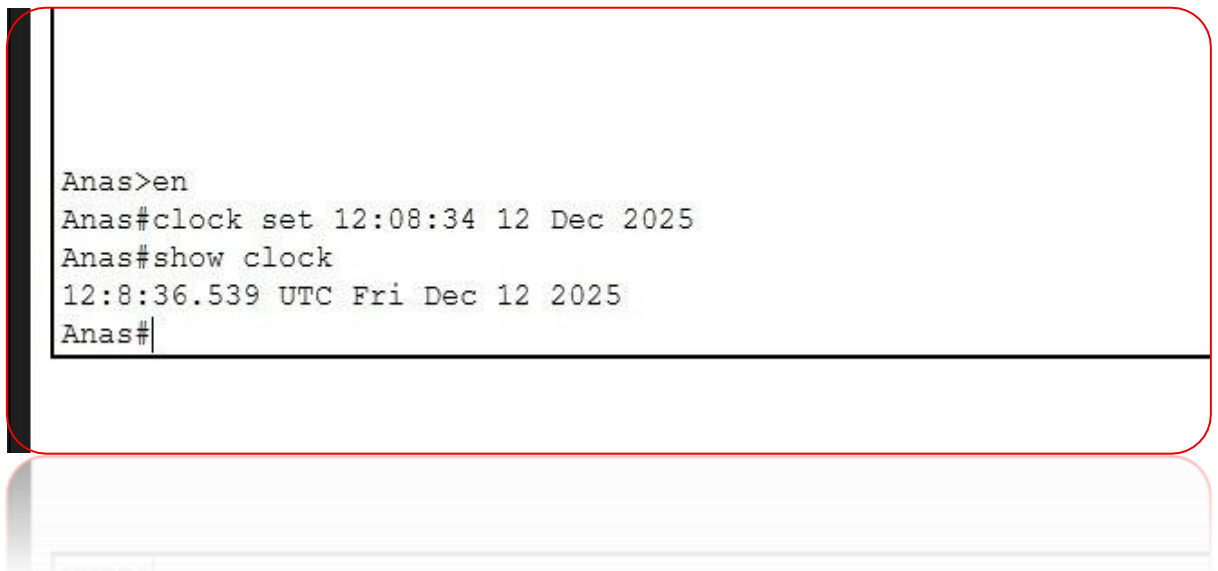
4.2.1 Hostname Configuration:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host Anas
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console
|
```

management.

⑨ This screenshot shows the configuration of a hostname on the router using the Cisco IOS command line, which helps in uniquely identifying the device and simplifying network administration.

4.2.2 Time and Date Configuration:

A screenshot of a Cisco IOS command-line interface. The terminal window has a black background with white text. The commands and their outputs are as follows:

```
Anas>en
Anas#clock set 12:08:34 12 Dec 2025
Anas#show clock
12:8:36.539 UTC Fri Dec 12 2025
Anas#
```

The screenshot is framed by a red border with rounded corners. The text is left-aligned and shows the transition from user EXEC mode to privileged EXEC mode, followed by setting the system clock and verifying it.

⑨ This screenshot shows the configuration of system time and date on the router using the clock set command, which is important for accurate logging and network monitoring.

4.2.3 Console and VTY Line Password Configuration:

```
Anas>en
Anas#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Anas(config)#line vty 04
Anas(config-line)#password hassan
Anas(config-line)#login
Anas(config-line)#exit
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console
```

⑨ This screenshot shows the configuration of VTY line password on the router to secure remote access and ensure that only authorized users can log in to the device

4.2.4 Console Line and Local User Configuration:

```

Anas>en
Anas#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Anas(config)#line vty 04
Anas(config-line)#password hassan
Anas(config-line)#login
Anas(config-line)#exit
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console

Anas#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Anas(config)#line console 0
Anas(config-line)#login local
Anas(config-line)#exit
Anas(config)#username Qamar password BU
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console

```

- ⑨ This screenshot shows the configuration of console line login using local user authentication, along with the creation of a local username and password to enhance device access security.

4.2.5 Enable Password and Enable Secret Configuration

```

Anas#config t
Enter configuration commands, one per line. End with CNTL/Z.
Anas(config)#enable password farooq
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console

Anas#config t
Enter configuration commands, one per line. End with CNTL/Z.
Anas(config)#enable secret mahar
Anas(config)#exit
Anas#
%SYS-5-CONFIG_I: Configured from console by console

Anas#exit

```

- ⑨ This screenshot shows the configuration of the enable password and enable secret on the router to protect privileged EXEC mode and ensure secure administrative access.

4.2.6 User Access Verification

```
User Access Verification
Username: Qamar
Password:
Anas>en
Password:
Password:
Anas#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Anas(config)#
```

⑨ This screenshot verifies user authentication by successfully logging into the router using the configured username and password, confirming that security settings are functioning correctly.

4.3 VLAN Configuration:

4.3.1 VLAN Creation and Access Port Assignment (IT Department)

```

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT-01
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT-02
Switch(config-vlan)#interface fa0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#
Switch(config-if)#exit
Switch(config)#

```

9 This screenshot shows the creation of VLAN 10 and VLAN 20 for the IT department and the assignment of access ports to these VLANs in order to separate departmental traffic.

4.3.2 VLAN Verification using show vlan Command:

```
Switch>show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT-01	active	Fa0/1
20	IT-02	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

--More--

1003	token	101003	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1	enet	100001	1500	-	-	-	-	-	0	0

9 This screenshot verifies the successful creation of VLANs and confirms that the correct switch ports are assigned to their respective VLANs using the show vlan command

4.3.3 VLAN Configuration on IT Department Switch (Additional Switch)

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL-Z
Switch(config)#vlan 10
Switch(config-vlan)#name IT-03
Switch(config-vlan)#interface fa0/4
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT-04
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

```
Switch(config)#
Switch(config-if)#exit
Switch(config-if)#switchport mode trunk
```

- ⑨ This screenshot shows the configuration of VLAN 10 and VLAN 20 on another IT department switch, including access port assignment and trunk port configuration for interswitch communication.

4.3.4 VLAN Configuration for Management Department:

```

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 50
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport access vlan 50
Switch(config-if)#name Management-01
Switch(config-if)#^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#vlan 50
Switch(config-vlan)#name Management-01
Switch(config-vlan)#exit
Switch(config)#vlan 60
Switch(config-vlan)#name Management-02
Switch(config-vlan)#interface fa0/3
Switch(config-if)#switchport access vlan 60
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#

```

```

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch(config-if)#

```

- ⑨ This screenshot shows the creation of VLAN 50 and VLAN 60 for the Management department, along with access port assignment and trunk port configuration to enable interswitch communication.

4.3.5 VLAN Configuration on Additional Management Switch:

```

Management>en
Management#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Management(config)#vlan 50
Management(config-vlan)#name Management-03
Management(config-vlan)#interface fa0/4
Management(config-if)#switchport access vlan 60
% Access VLAN does not exist. Creating vlan 60
Management(config-if)#exit
Management(config)#vlan 60
Management(config-vlan)#name Management-04
Management(config-vlan)#interface fa0/1
Management(config-if)#switchport access vlan 60
Management(config-if)#exit
Management(config)#vlan 50
Management(config-vlan)#name Management-03
Management(config-vlan)#interface fa0/4
Management(config-if)#switchport access vlan 50
Management(config-if)#exit
Management(config)#interface fa0/2
Management(config-if)#switchport mode trunk
Management(config-if)#exit
Management(config)#exit
Management#
%SYS-5-CONFIG_I: Configured from console by console

```

- ⑨ This screenshot shows the configuration of VLAN 50 and VLAN 60 on an additional Management department switch, including access port assignment and trunk port configuration for inter-switch connectivity.

4.3.6 VLAN Verification for Management Department:

```
Management>show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
50	Management-03	active	Fa0/4
60	Management-04	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
50	enet	100050	1500	-	-	-	-	-	0	0
60	enet	100060	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

9 This screenshot verifies the successful configuration of Management department VLANs and confirms that the correct switch ports are assigned to their respective VLANs using the show vlan command.

4.3.7 VLAN Configuration for Finance Department:

```

Switch>
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Finance
Finance(config)#vlan 90
Finance(config-vlan)#name Finance-01
Finance(config-vlan)#interface fa0/3
Finance(config-if)#switchport access vlan 90
Finance(config-if)#exit
Finance(config)#vlan 100
Finance(config-vlan)#name Finance-02
Finance(config-vlan)#interface fa0/31
      ^
% Invalid input detected at '^' marker.

Finance(config-vlan)#interface fa0/1
Finance(config-if)#switchport access vlan 100
Finance(config-if)#exit
Finance(config)#interface fa0/2
Finance(config-if)#switchport mode trunk

Finance(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

```

⑨ This screenshot shows the creation of VLAN 90 and VLAN 100 for the Finance department, including access port assignment and trunk port configuration to enable communication with other switches.

4.3.8 VLAN Verification for Finance Department:

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
90	Finance-01	active	Fa0/3
100	Finance-02	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
90	enet	100090	1500	-	-	-	-	-	0	0
100	enet	100100	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

--More--										
1003	enet	1001003	1500	-	-	-	-	-	0	0
100	enet	100100	1500	-	-	-	-	-	0	0
90	enet	100090	1500	-	-	-	-	-	0	0
1	enet	100001	1500	-	-	-	-	-	0	0

⑨ This screenshot verifies the successful configuration of Finance department VLANs and confirms that the correct switch ports are assigned to VLAN 90 and VLAN 100 using the show vlan command.

4.3.9 VLAN Configuration on Additional Finance Switch:

```
Finance-F2>en
Finance-F2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Finance-F2(config)#vlan 90
Finance-F2(config-vlan)#name Finance-03
Finance-F2(config-vlan)#interface fa0/4
Finance-F2(config-if)#switchport access vlan 90
Finance-F2(config-if)#exit
Finance-F2(config)#vlan 100
Finance-F2(config-vlan)#name Finance-04
Finance-F2(config-vlan)#interface fa0/1
Finance-F2(config-if)#switchport access vlan 100
Finance-F2(config-if)#exit
Finance-F2(config)#interface fa0/2
Finance-F2(config-if)#switchport mode trunk
Finance-F2(config-if)#exit
Finance-F2(config)#
```

```
Finance-F2(config)#
Finance-F2(config-if)#exit
Finance-F2(config-if)#switchport mode trunk
Finance-F2(config-if)#
```

- ⑨ This screenshot shows the configuration of VLAN 90 and VLAN 100 on an additional Finance department switch, including access port assignment and trunk port configuration for inter-switch communication.

4.3.10 VLAN Verification for Additional Finance Switch:

Finance-F2>show vlan

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
90	Finance-03	active	Fa0/4
100	Finance-04	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
90	enet	100090	1500	-	-	-	-	-	0	0
100	enet	100100	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

⑨ This screenshot verifies the successful configuration of VLAN 90 and VLAN 100 on the additional Finance department switch and confirms correct port-to-VLAN mapping using the show vlan command.

4.3.11 VLAN Configuration for Computer Science Department:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname ComputerScience-C2
ComputerScience-C2(config)#vlan 130
ComputerScience-C2(config-vlan)#name ComputerScience-03
ComputerScience-C2(config-vlan)#interface fa0/4
      ^
% Invalid input detected at '^' marker.

ComputerScience-C2(config-vlan)#interface fa0/4
ComputerScience-C2(config-if)#switchport access vlan 130
ComputerScience-C2(config-if)#exit
ComputerScience-C2(config)#vlan 140
ComputerScience-C2(config-vlan)#name ComputerScience-04
ComputerScience-C2(config-vlan)#interface fa0/1
ComputerScience-C2(config-if)#switchport access vlan 140
ComputerScience-C2(config-if)#exit
ComputerScience-C2(config)#interface fa0/2
ComputerScience-C2(config-if)#switchport mode trunk

ComputerScience-C2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

```

- ⑨ This screenshot shows the configuration of VLAN 130 and VLAN 140 for the Computer Science department, including access port assignment and trunk port configuration to enable inter-switch connectivity.

4.3.12 VLAN Configuration on Computer Science Main Switch:

```

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname ComputerScience
ComputerScience(config)#vlan 130
ComputerScience(config-vlan)#name ComputerScience-01
ComputerScience(config-vlan)#interface fa0/3
ComputerScience(config-if)#switchport access vlan 130
ComputerScience(config-if)#exit
ComputerScience(config)#vlan 140
ComputerScience(config-vlan)#name ComputerScience-02
ComputerScience(config-vlan)#interface fa0/1
ComputerScience(config-if)#switchport access vlan 140
ComputerScience(config-if)#exit
ComputerScience(config)#interface fa0/2
ComputerScience(config-if)#switchport mode trunk
ComputerScience(config-if)#exit
ComputerScience(config)#exit
ComputerScience#
%SYS-5-CONFIG_I: Configured from console by console

```

```

%SYS-5-CONFIG_I: Configured from console by console
ComputerScience#
ComputerScience(config)#exit
ComputerScience(config-if)#exit
ComputerScience(config-if)#exit
ComputerScience(config-if)#exit

```

- ⑨ This screenshot shows the configuration of VLAN 130 and VLAN 140 on the main Computer Science department switch, including access port assignment and trunk port configuration for inter-switch communication.

4.3.13 VLAN Verification for Computer Science Department:

```
ComputerScience>show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
130	ComputerScience-01	active	Fa0/3
140	ComputerScience-02	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
130	enet	100130	1500	-	-	-	-	-	0	0
140	enet	100140	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More-- |

--More--

1003	enet	1001003	1500	-	-	-	-	-	0	0
140	enet	100140	1500	-	-	-	-	-	0	0
130	enet	100130	1500	-	-	-	-	-	0	0
1	enet	100001	1500	-	-	-	-	-	0	0

⑨ This screenshot verifies the successful configuration of Computer Science department VLANs and confirms correct port assignment for VLAN 130 and VLAN 140 using the show vlan command.

4.4 DHCP Configuration for IT Department:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host IT
IT(config)#ip dhcp excluded-address 40.0.0.1
IT(config)#ip dhcp pool IT-POOL
IT(dhcp-config)#network 40.0.0.0 255.0.0.0
IT(dhcp-config)#default-router 40.0.0.1
IT(dhcp-config)#dns-server 8.8.8.8
IT(dhcp-config)#exit
IT(config)#ip dhcp pool 41.0.0.1
IT(dhcp-config)#network 41.0.0.0 255.0.0.0
IT(dhcp-config)#default-router 41.0.0.1
IT(dhcp-config)#dns-server 8.8.8.8
IT(dhcp-config)#exit
IT(config)#
```

- ⑨ This screenshot shows the configuration of DHCP pools on the router for the IT department, enabling automatic IP address allocation along with default gateway and DNS server settings.

4.5 Configuration Backup Using TFTP Server:

```

ANAS>en
ANAS#wr
Building configuration...
[OK]
ANAS#copy startup-config tftp:
Address or name of remote host []? 192.168.7.2
Destination filename [ANAS-config]? Hassan

Writing startup-config...!!
[OK - 1422 bytes]

1422 bytes copied in 0 secs
ANAS#copy startup-config tftp:
Address or name of remote host []? 192.168.7.2
Destination filename [ANAS-config]? IT

Writing startup-config...!!
[OK - 1422 bytes]

1422 bytes copied in 0 secs
ANAS#copy startup-config tftp:
Address or name of remote host []? 192.168.7.2
Destination filename [ANAS-config]? Finance

Writing startup-config...!!
[OK - 1422 bytes]

1422 bytes copied in 0 secs
ANAS#copy startup-config tftp:
Address or name of remote host []? 192.168.7.2
Destination filename [ANAS-config]? Management

Writing startup-config...!!
[OK - 1422 bytes]

1422 bytes copied in 0 secs
ANAS#copy startup-config tftp:
Address or name of remote host []? 192.168.7.2
Destination filename [ANAS-config]? ComputerScience

```

```

Destination filename [ANAS-config]? ComputerScience
Address or name of remote host []? 192.168.7.2
ANAS#copy startup-config tftp:
1422 bytes copied in 0 secs

[OK - 1422 bytes]
Writing startup-config...!!

Destination filename [ANAS-config]? Management
Address or name of remote host []? 192.168.7.2

```

- ⑨ This screenshot shows the successful backup of router startup configurations to a centralized TFTP server, ensuring configuration recovery and network reliability in case of device failure.

4.6 DHCP Verification using show ip dhcp pool

Command:

```
Router>
Router>show ip dhcp pool

Pool VLAN90 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 2
  Excluded addresses             : 2
  Pending event                  : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.90.1      192.168.90.1 - 192.168.90.254    2 / 2 / 254

Pool VLAN910 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 511
  Leased addresses                : 0
  Excluded addresses             : 2
  Pending event                  : none

  0 subnet is currently in the pool

Pool VLAN100 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 2
  Excluded addresses             : 2
  Pending event                  : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.100.1      192.168.100.1 - 192.168.100.254    2 / 2 / 254
Router>
```

```
Router>
192.168.100.1      192.168.100.1 - 192.168.100.254    2 / 2 / 254
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 2
  Excluded addresses             : 2
  Pending event                  : none
Router>
```

- ⑨ This screenshot verifies the DHCP configuration by displaying active DHCP pools and leased IP addresses, confirming that devices are successfully receiving IP addresses dynamically.

4.7 DHCP Verification for Computer Science

Department:

```
Router>show ip dhcp pool

Pool VLAN130 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 4
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.130.1      192.168.130.1 - 192.168.130.254  2 / 4 / 254

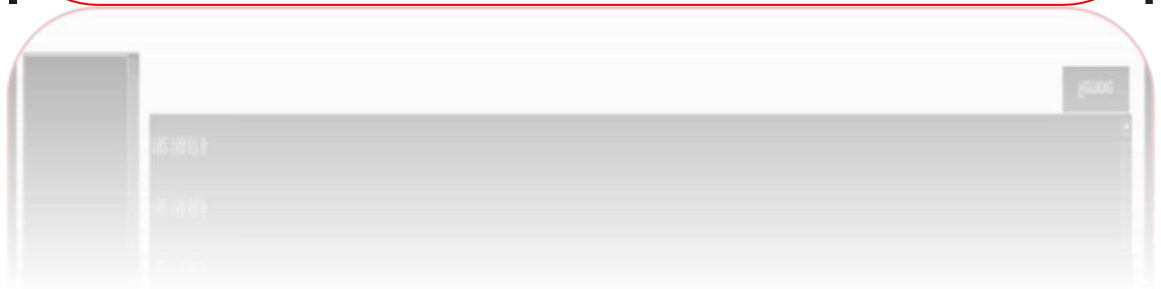
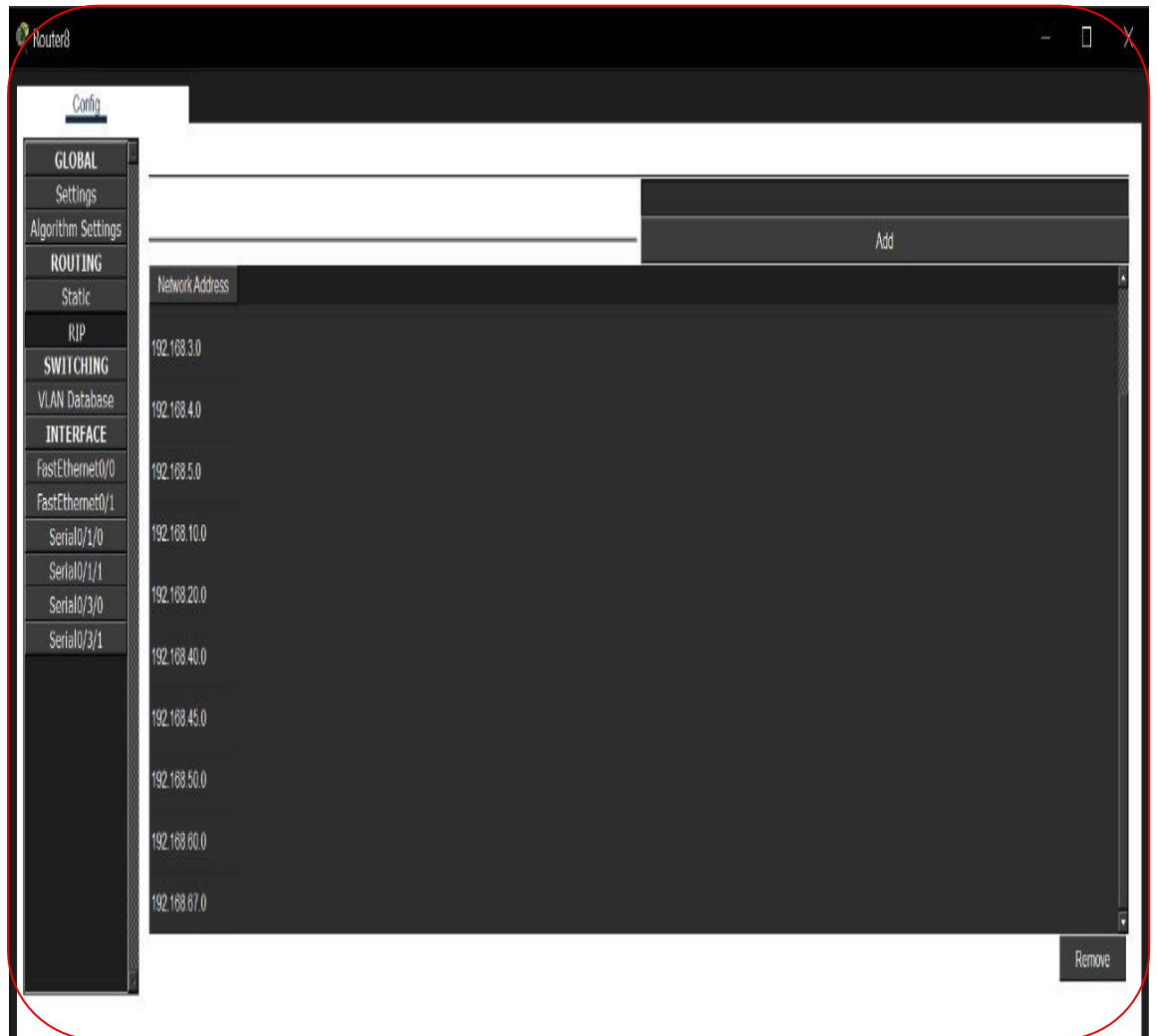
Pool VLAN140 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 4
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.140.1      192.168.140.1 - 192.168.140.254  2 / 4 / 254
```

- ⑨ This screenshot verifies the DHCP configuration for the Computer Science department by showing active DHCP pools

and leased IP addresses, confirming successful dynamic IP allocation.

4.8 RIP Routing Configuration:



⑨ This screenshot shows the configuration of the RIP routing protocol on the router, where multiple network addresses are added to enable dynamic routing and communication between different VLANs and network segments.

4.9 Inter-VLAN Connectivity Verification (Ping Test):

⑨ This screenshot shows successful ICMP communication between devices across different VLANs, confirming proper



The screenshot displays a network simulation interface. At the top right, there are two tabs: 'Realtime' (active) and 'Simulation'. On the left side, there is a control panel with an information icon, a dropdown menu set to 'Scenario 0', and buttons for 'New', 'Delete', and 'Toggle PDU List Window'. The main area contains a table with the following data:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Server0	ICMP	Blue	0.000	N	0	(edit)	(delete)
	Successful	PC16	PC1	ICMP	Green	0.000	N	1	(edit)	(delete)
	Successful	PC0	PC17	ICMP	Orange	0.000	N	2	(edit)	(delete)
	Successful	PC24	Server0	ICMP	Brown	0.000	N	3	(edit)	(delete)

inter-VLAN routing and overall network connectivity.

4.10 End Device IP Assignment Verification:



- ⑨ This screenshot shows that the end device successfully obtained an IP address, subnet mask, default gateway, and DNS

server automatically through DHCP, confirming correct DHCP configuration

5. Results and Validation:

⑨ **IP Assignment:** DHCP successfully assigned IP addresses to end devices across all configured VLANs.

⑨ **Routing:** RIP routing protocol was successfully configured, and routing information was verified using routing tables to ensure proper network connectivity.

⑨ **VLAN Communication:** Inter-VLAN communication through router-based routing was successfully achieved, allowing devices from different VLANs to communicate as required.

⑨ **Backup:** Device configuration files were successfully backed up to a centralized TFTP server for recovery purposes. ⑨ **Security:** Device access was secured using console, VTY, and enable passwords, ensuring restricted and authorized access to network devices.

6. Conclusion and Recommendations:

This project demonstrates the successful implementation of a secure and scalable enterprise network using Cisco Packet Tracer. The use of VLANs ensured logical segmentation and improved network performance, while DHCP simplified IP address management. RIP routing enabled effective communication between different network segments, and configuration backups enhanced network reliability. Overall, the project meets all CCN lab requirements and provides practical exposure to real-world networking concepts. Future enhancements may include the integration of wireless networks,

implementation of redundancy protocols, and advanced network monitoring solutions.

7. Appendix: Network Diagram:

This section includes the complete network topology diagram illustrating the interconnection of routers, switches, VLANs, and end devices used in the project.

